

E055

Project Scoping and Scheduling

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A project manager at a mid-size hospital, who has led four prior renovations, is building a WBS for a new patient-portal launch. The sponsor wants the portal live before flu season. Which practice follows the 100% rule?

- A. Include only the major deliverables and track minor work informally.
- B. Every work package must take exactly 8 to 80 hours or the WBS is invalid.
- C. List every task in date order so the WBS doubles as the schedule.
- D. Make sure the elements under each parent add up to all of that parent's scope, with nothing extra and nothing missing.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

On a software project, testers can begin once coding has begun, working on the first finished modules while coders keep going. The team uses a cloud scheduling tool licensed last spring and meets on Tuesdays. Which dependency links coding to testing?

- A. Finish-to-finish.
- B. Start-to-start.
- C. Start-to-finish.
- D. Finish-to-start.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A training-center build has these activities: A (3 days) starts the project; B (4 days) and C (6 days) both follow A; E (5 days) can start on day one; D (2 days) follows B, C, and E. Activity B uses the most expensive contractor, and E has the most team members. What is the project's minimum duration?

- A. 7 days
- B. 9 days
- C. 11 days
- D. 20 days

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

A project has two paths: Path 1 is 14 days and Path 2 is 10 days. An activity that is only on Path 2 slips by 3 days. What is the new project duration?

- A.** 13 days
- B.** 17 days
- C.** 14 days
- D.** 27 days

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

Activity F has an early start of day 5, early finish of day 9, late start of day 8, and late finish of day 12. The activity is handled by a vendor in another time zone that bills weekly. What is F's total float?

- A.** 3 days
- B.** 12 days
- C.** 4 days
- D.** 7 days

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

An activity has total float of 5 days, an early start of day 10, and a duration of 4 days. What is its late finish?

- A.** Day 19
- B.** Day 9
- C.** Day 15
- D.** Day 14

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

Two critical activities both need the only certified electrician, who is also the company's safety trainer, during the same week. The project manager applies resource leveling. The electrician's union contract renews next year. What is the most likely effect?

- A. The finish date cannot change, because leveling only uses float.
- B. The project finish date may move later, because leveling delays one critical activity to fix the over-allocation.
- C. The union renewal next year will extend the schedule.
- D. The finish date moves earlier, because work is spread more evenly.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A project must recover 2 days. Critical activity G can be shortened up to 2 days at \$2,000 per day; critical activity H can be shortened up to 1 day at \$1,500 per day. Activity J, which is not on the critical path, can be shortened up to 2 days at \$1,200 per day. No other path becomes critical. G's vendor has the best quality rating in the region. What is the lowest cost to recover the 2 days?

- A. \$4,000
- B. \$3,000
- C. \$3,500
- D. \$2,400

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

A project manager needs to shorten the schedule without adding cost and is willing to accept more rework risk. Which technique fits?

- A. Cutting scope without approval.
- B. Crashing.
- C. Resource leveling.
- D. Fast tracking.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

At month 4, a dashboard project has planned value of \$80,000, earned value of \$68,000, and actual cost of \$72,000. The team moved offices in month 3. What is the schedule performance index?

- A.** 1.18
- B.** 0.85
- C.** -\$12,000
- D.** 0.94

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

A project's SPI is 1.10 and its planned value to date is \$50,000. What is its earned value?

- A.** \$50,000
- B.** \$45,000
- C.** \$45,455
- D.** \$55,000

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

Mid-project, a task is running 5 days late because a supplier shipped late. The project manager, who is PMP-certified, is asked to make the schedule look on track before the steering committee's quarterly review. What is the appropriate response?

- A.** Report the variance, take corrective action such as fast tracking or adding resources, and change the baseline only through approved change control.
- B.** Wait to report the delay until after the quarterly review.
- C.** Do nothing; small variances always correct themselves.
- D.** Update the baseline dates so the variance disappears.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A project manager at a mid-size hospital, ~~who has led four prior renovations~~, is building a WBS for a new patient-portal launch. ~~The sponsor wants the portal live before flu season~~. Which practice follows the 100% rule?

Target word: the 100% rule **Decides:** Children must sum to all of the parent's scope, no more and no less.

A	Dropped element	Leaving out minor work breaks the 100% rule.
B	Absolute language	The 8/80 guideline is a rule of thumb, not a validity test.
C	Adjacent category	Sequencing and dates belong to the schedule, not the WBS.
D	Correct	That is the 100% rule.

2. Answer B Concept 2

On a software project, testers can begin once coding has begun, working on the first finished modules while coders keep going. ~~The team uses a cloud scheduling tool licensed last spring and meets on Tuesdays~~. Which dependency links coding to testing?

Target word: dependency type **Decides:** Testing may start once coding has started.

A	Adjacent category	Ties two finishes; the stem is about when testing can begin.
B	Correct	The successor's start depends on the predecessor's start.
C	Inverted operation	Would mean coding cannot finish until testing starts, a reversed and rare link.
D	Confusion set	The default type, but testing does not wait for all coding to finish.

3. Answer C Concept 3

A training-center build has these activities: A (3 days) starts the project; B (4 days) and C (6 days) both follow A; E (5 days) can start on day one; D (2 days) follows B, C, and E. ~~Activity B uses the most expensive contractor, and E has the most team members~~. What is the project's minimum duration?

Target word: minimum project duration **Decides:** Longest path: A-C-D = 3 + 6 + 2 = 11.

A	Camouflage	The path through E, the activity with the most people; staff count does not decide length.
B	Confusion set	The A-B-D path, which has 2 days of float.
C	Correct	Paths: A-B-D = 9, A-C-D = 11, E-D = 7. The longest is 11.
D	Inverted operation	Adds every activity as if all ran in one line.

4. Answer C Concept 3 · transfer

A project has two paths: Path 1 is 14 days and Path 2 is 10 days. An activity that is only on Path 2 slips by 3 days. What is the new project duration?

Target word: effect of a non-critical slip **Decides:** Path 2 becomes 13 days, still shorter than Path 1.

A	Dropped element	The new length of Path 2, not the project.
B	Adjacent category	Adds the slip to the project as if the activity were critical.
C	Correct	Path 2 grows to 13, still under 14; the critical path is unchanged.
D	Inverted operation	Adds the two paths plus the slip instead of taking the longest path.

5. Answer A Concept 4

Activity F has an early start of day 5, early finish of day 9, late start of day 8, and late finish of day 12. ~~The activity is handled by a vendor in another time zone that bills weekly.~~ What is F's total float?

Target word: total float **Decides:** $LS - ES = 8 - 5$.

A	Correct	$LS - ES = 8 - 5 = 3$ (also $LF - EF = 12 - 9 = 3$).
B	Wrong stem number	The late finish date, not a float amount.
C	Confusion set	$EF - ES$ is the duration, not float.
D	Dropped element	$LF - ES$ leaves the duration inside the figure.

6. Answer A Concept 4 • transfer

An activity has total float of 5 days, an early start of day 10, and a duration of 4 days. What is its late finish?

Target word: solve for late finish **Decides:** $LS = ES + \text{float} = 15$; $LF = LS + \text{duration}$.

A	Correct	$LS = 10 + 5 = 15$; $LF = 15 + 4 = 19$.
B	Inverted operation	Subtracts float instead of adding it.
C	Dropped element	That is the late start; the duration still has to be added.
D	Adjacent category	That is the early finish ($10 + 4$).

7. Answer B Concept 5

Two critical activities both need the only certified electrician, ~~who is also the company's safety trainer~~, during the same week. The project manager applies resource leveling. ~~The electrician's union contract renews next year.~~ What is the most likely effect?

Target word: effect of leveling on critical work **Decides:** Both activities are critical, so delaying one moves the end date.

A	Confusion set	That describes resource smoothing, not leveling.
B	Correct	Leveling respects resource limits even if the critical path grows.
C	Camouflage	The contract date does not affect this week's conflict.
D	Inverted operation	Spreading work to fit one person does not shorten the schedule.

8. Answer C Concept 6

A project must recover 2 days. Critical activity G can be shortened up to 2 days at \$2,000 per day; critical activity H can be shortened up to 1 day at \$1,500 per day. Activity J, which is not on the critical path, can be shortened up to 2 days at \$1,200 per day. No other path becomes critical. ~~G's vendor has the best quality rating in the region.~~ What is the lowest cost to recover the 2 days?

Target word: cheapest crash plan **Decides:** Crash critical activities only, cheapest per day first, within limits.

A	Camouflage	Crashes G for both days because of its vendor rating; rating does not decide cost.
B	Dropped element	Crashes H for 2 days, ignoring its 1-day limit.
C	Correct	H for 1 day (\$1,500) plus G for 1 day (\$2,000) = \$3,500.
D	Adjacent category	Crashing J, a non-critical activity, does not shorten the project.

9. Answer D Concept 6 • transfer

A project manager needs to shorten the schedule without adding cost and is willing to accept more rework risk. Which technique fits?

Target word: pick the technique from its cost/risk profile **Decides:** No added cost, added risk = overlapping work.

A	Adjacent category	Scope reduction requires change control and is not a schedule-compression technique.
B	Confusion set	Crashing adds resources and cost.
C	Inverted operation	Leveling tends to lengthen, not shorten, the schedule.
D	Correct	Overlapping activities saves time without buying resources, but raises rework risk.

10. Answer B Concept 7

At month 4, a dashboard project has planned value of \$80,000, earned value of \$68,000, and actual cost of \$72,000. ~~The team moved offices in month 3.~~ What is the schedule performance index?

Target word: SPI **Decides:** $SPI = EV / PV$.

A	Inverted operation	PV / EV flips the ratio.
B	Correct	$\$68,000 / \$80,000 = 0.85$, behind schedule.
C	Adjacent category	That is schedule variance (EV – PV), not the index.
D	Confusion set	EV / AC is the cost performance index, not SPI.

11. Answer D Concept 7 • transfer

A project's SPI is 1.10 and its planned value to date is \$50,000. What is its earned value?

Target word: solve for EV **Decides:** $EV = SPI \times PV$.

A	Wrong stem number	That is the planned value.
B	Confusion set	Reads an SPI above 1 as behind schedule.
C	Inverted operation	Divides PV by SPI instead of multiplying.
D	Correct	$1.10 \times \$50,000 = \$55,000$, ahead of schedule.

12. Answer A Concept 8

Mid-project, a task is running 5 days late because a supplier shipped late. The project manager, ~~who is PMP-certified~~, is asked to make the schedule look on track before the steering committee's quarterly review. What is the appropriate response?

Target word: handling a schedule variance **Decides:** Baseline changes need approved change control; variances get corrective action.

A	Correct	Keeps reporting honest and uses the formal process for any baseline change.
B	Dropped element	Drops timely, honest reporting of the variance.
C	Absolute language	Variances do not reliably self-correct; they need action.
D	Confusion set	Re-baselining without approval hides the variance instead of correcting it.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	6	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	7	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	2	
Confusion set	Every option is real; the test is exact role-matching.	8	
Camouflage	Answers a stem fact that was never the question.	3	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	WBS and the 100% rule	A WBS breaks all project scope into deliverable-based pieces; each level's children add up to exactly 100% of the parent's scope.	Treating the WBS as a timeline of tasks, or leaving out minor work.	"decompose," "work packages," "all of the scope."
2	Dependency types	Finish-to-start is the default; start-to-start lets a successor begin once the predecessor has begun; finish-to-finish ties two finishes; start-to-finish is rare.	Calling every link finish-to-start.	"can begin once ... has begun" or "cannot finish until."
3	Critical path	The critical path is the longest path through the network and sets the shortest possible project duration.	Picking the shortest path or adding every activity together.	A network with parallel paths and "minimum duration."
4	Total float	Total float = LS – ES (or LF – EF); critical activities have zero float, and a slip beyond float delays the project.	Confusing float with duration.	ES, EF, LS, LF values in the stem.
5	Resource leveling vs. smoothing	Leveling resolves over-allocation and can move the end date; smoothing only uses available float and keeps the end date.	Thinking leveling never changes the finish date.	One scarce person needed by two critical tasks at once.
6	Schedule compression	Crashing adds resources to critical activities at extra cost (cheapest per day first); fast tracking overlaps activities at extra risk.	Crashing non-critical activities, or thinking fast tracking adds cost.	"Recover days" with a cost per day, or "overlap."
7	Schedule variance and SPI	$SV = EV - PV$ and $SPI = EV / PV$; SPI below 1 means behind schedule.	Using AC in schedule measures, or inverting the ratio.	PV, EV, and AC given together with a schedule question.
8	Corrective action vs. re-baselining	Variances are reported and addressed with corrective action; the baseline changes only through approved change control.	Moving the baseline to make a variance disappear.	Pressure to make a late schedule look on track.

Course scope (public description): In Project Scoping and Scheduling, learners will develop their skills in creating a work breakdown structure (WBS) with defined project activities and interdependencies aligned with project objectives. Learners will master the development of project schedules using critical path method, resource leveling, and scheduling software tools to optimize project timelines and resource allocation. Additionally, the learner will gain expertise in monitoring project schedules, identifying variances, and implementing corrective actions to ensure project progress aligns with the planned schedule, enhancing their ability to effectively manage project timelines and deliver successful project outcomes.